Data applications: Type 2 diabetes trio data

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Purpose

This document reconstructs the case-parent trio analysis used in the manuscript. It explains the conditional-likelihood formulation, shows how Mendelian transmission probabilities enter as offsets in conditional logistic regression, and fits an additive per-allele model for the Z+2 allele at the *GCK1* microsatellite locus.

The reconstructed data are based on the counts reported by Frayling et al. (1999). The affected child is treated as the case, and the genotypes that could have been transmitted by the same parents are treated as matched pseudo-controls. Parental mating type defines the matched set.

Data and notation

Let

- $G \in \{0, 1, 2\}$ denote the number of copies of the Z+2 allele carried by the child;
- $G_p = (g_{p1}, g_{p2})$ denote the parental Z+2 allele counts;
- $D = 1$ identify the observed affected child and $D = 0$ identify a pseudo-control genotype.

The original draft described the genetic variants as SNPs. That is not correct for this application: *GCK1* is a microsatellite locus, and Z+2 is one of its alleles.

The reported affected-child counts are:

```r
affected_counts <- data.frame(
  mating = c("0x1", "0x1", "1x1", "1x1"),
  observed_g = c(0, 1, 0, 1),
  n = c(10, 15, 1, 1)
)

affected_counts
```

```
##   mating observed_g  n
## 1    0x1          0 10
## 2    0x1          1 15
## 3    1x1          0  1
## 4    1x1          1  1
```

Conditional likelihood for trio data

Conditional on parental mating type G_p, the probability that the affected child has genotype g is proportional to

$$\exp\{z(g)^T \beta\} P(G = g \mid G_p),$$

where $P(G = g \mid G_p)$ is the Mendelian transmission probability. Thus the likelihood contribution of one trio is

$$\frac{\exp\{z(g)^T\beta\}P(G=g \mid G_p)}{\sum_{g^*}\exp\{z(g^*)^T\beta\}P(G=g^* \mid G_p)}.$$

This has the form of a conditional logistic likelihood with linear predictor

$$z(g)^T\beta + \log P(G=g \mid G_p).$$

Therefore, the logarithm of the Mendelian transmission probability enters the conditional logistic regression as an offset.

Because an additive constant within a matched set cancels from the conditional likelihood, one may use either the exact log probabilities or any proportional weights. For a 1×1 mating, for example,

$$P(G=0 \mid G_p) = \frac{1}{4}, \qquad P(G=1 \mid G_p) = \frac{1}{2}, \qquad P(G=2 \mid G_p) = \frac{1}{4}.$$

The exact offsets are $\{\log(1/4), \log(1/2), \log(1/4)\}$. Subtracting the common value $\log(1/4)$ gives the equivalent offsets

$$\{0, \log 2, 0\}.$$

Additive per-allele model

The manuscript reports a single genotype relative risk for each additional copy of the Z+2 allele. The corresponding model is

$$z(g) = g$$

and

$$\mathrm{GRR} = \exp(\beta).$$

This is different from a two-parameter model with separate genotype effects. If one writes

$$z(g) = \begin{cases} (0,0), & g=0, \\ (1,0), & g=1, \\ (1,1), & g=2, \end{cases}$$

then the parameters are log relative-risk increments:

$$\exp(\beta_1) = \frac{P(D=1 \mid G=1)}{P(D=1 \mid G=0)}, \qquad \exp(\beta_2) = \frac{P(D=1 \mid G=2)}{P(D=1 \mid G=1)}.$$

The β_j themselves are log-GRRs, not GRRs. In these reconstructed data, no affected child has genotype $G=2$, so the two-parameter model has no information about β_2. The single-parameter additive model is therefore the model used below.

Reconstructing the matched pseudo-control data

For a 0×1 mating, the possible offspring genotypes are 0 and 1, each with probability 1/2.

For a 1×1 mating, the possible offspring genotypes are 0, 1, and 2, with probabilities 1/4, 1/2, and 1/4, respectively.

Each observed affected child defines one matched set. The observed genotype is coded as the case, and the other possible offspring genotypes are coded as pseudo-controls.

```r
make_set <- function(stratum, mating, observed_g) {
  if (mating == "0x1") {
    # Mendelian probabilities: P(G=0)=1/2, P(G=1)=1/2
    g <- c(0, 1)

  } else if (mating == "1x1") {
    # Mendelian probabilities: P(G=0)=1/4, P(G=1)=1/2, P(G=2)=1/4
    # Represent these probabilities by multiplicities 1:2:1.
    g <- c(0, 1, 1, 2)

  } else {
    stop("Unsupported mating type: ", mating)
  }

  case <- integer(length(g))

  # Mark exactly one copy of the observed genotype as the affected child.
  observed_rows <- which(g == observed_g)

  if (length(observed_rows) == 0L) {
    stop(
      "Observed genotype ", observed_g,
      " is incompatible with mating type ", mating, "."
    )
  }

  case[observed_rows[1L]] <- 1L

  data.frame(
    stratum = stratum,
    mating = mating,
    g = g,
    case = case
  )
}

sets <- vector("list", sum(affected_counts$n))
s <- 1L

for (i in seq_len(nrow(affected_counts))) {
  for (j in seq_len(affected_counts$n[i])) {
    sets[[s]] <- make_set(
      stratum = s,
      mating = affected_counts$mating[i],
      observed_g = affected_counts$observed_g[i]
    )
```

```
    s <- s + 1L
  }
}

trio_dat <- do.call(rbind, sets)
row.names(trio_dat) <- NULL

head(trio_dat, 12)
```

```
##    stratum mating g case
## 1        1    0x1 0    1
## 2        1    0x1 1    0
## 3        2    0x1 0    1
## 4        2    0x1 1    0
## 5        3    0x1 0    1
## 6        3    0x1 1    0
## 7        4    0x1 0    1
## 8        4    0x1 1    0
## 9        5    0x1 0    1
## 10       5    0x1 1    0
## 11       6    0x1 0    1
## 12       6    0x1 1    0
```

Basic checks:

```
stopifnot(all(tapply(trio_dat$case, trio_dat$stratum, sum) == 1))
stopifnot(nlevels(factor(trio_dat$stratum)) == 27)
stopifnot(nrow(trio_dat) == 58)

with(trio_dat, table(mating, g, case))
```

```
## , , case = 0
##
##       g
## mating  0  1  2
##    0x1 15 10  0
##    1x1  1  3  2
##
## , , case = 1
##
##       g
## mating  0  1  2
##    0x1 10 15  0
##    1x1  1  1  0
```

Fitting the additive conditional logistic regression

```
library(survival)

fit <- clogit(
  case ~ g + strata(stratum),
  data = trio_dat,
  method = "exact"
)
```

```
summary(fit)
```

```
## Call:
## coxph(formula = Surv(rep(1, 58L), case) ~ g + strata(stratum),
##     data = trio_dat, method = "exact")
##
##   n= 58, number of events= 27
##
##     coef exp(coef) se(coef)     z Pr(>|z|)
## g 0.2076    1.2308   0.3734 0.556    0.578
##
##   exp(coef) exp(-coef) lower .95 upper .95
## g     1.231     0.8125     0.592     2.559
##
## Concordance= 0.532  (se = 0.13 )
## Likelihood ratio test= 0.31  on 1 df,   p=0.6
## Wald test            = 0.31  on 1 df,   p=0.6
## Score (logrank) test = 0.31  on 1 df,   p=0.6
```

```
trio_dat$matchedset <- trio_dat$stratum
form <- formula(case ~ g)
#trio.aug0 <- augment.logFmatched(form,trio_dat,m=0) # doesn't work. Not set up for m=0
trio.aug0 <- trio_dat; trio.aug0$weights <- 1 # add weights manually
UCLR <- clogitlogF(form,trio.aug0,maxit=500)
UCLR
```

```
## Call:
## clogit(formula, dat, weights = dat$weights, method = "efron",
##     iter.max = maxit)
##
##     coef exp(coef) se(coef)     z     p
## g 0.2076    1.2308   0.3734 0.556 0.578
##
## Likelihood ratio test=0.31  on 1 df, p=0.5771
## n= 58, number of events= 27
```

```
exp(c(UCLR$ci.lower,UCLR$ci.upper))
```

```
##     lower     upper
## 0.5927131 2.6040805
```

```
FCLR <- clogitf(form,trio_dat,maxit=500)
FCLR
```

```
## coxphf(formula = newformula, data = data, pl = pl, alpha = alpha,
##     maxit = maxit, maxstep = 0.1, firth = firth, penalty = penalty)
## Model fitted by Penalized ML
## Confidence intervals and p-values by Profile Likelihood
##
##        coef  se(coef) exp(coef) lower 0.95 upper 0.95    Chisq         p
## g 0.2006707 0.3732617  1.222222  0.5959001   2.551013 0.300502 0.5835679
##
## Likelihood ratio test=0.300502 on 1 df, p=0.5835679, n=58
```

```
confint(FCLR)
```

```
##         [,1]      [,2]
```

```
## g -0.5176822 0.9364906
```

```r
exp(confint(FCLR))
```

```
##         [,1]     [,2]
## g 0.5959001 2.551013
```

The uninformative prior for DES and maternal smoking is log-F(1,1).

```r
trio.augU <- augment.logFmatched(form,trio_dat,m=1)
logFU <- clogitlogF(form,trio.augU)
logFU
```

```
## Call:
## clogit(formula, dat, weights = dat$weights, method = "efron",
##     iter.max = maxit)
##
##     coef exp(coef) se(coef) robust se     z     p
## g 0.2007    1.2222   0.3670    0.2727 0.736 0.462
##
## Likelihood ratio test=0.3  on 1 df, p=0.5836
## n= 62, number of events= 29
```

```r
confint(logFU)
```

```
##      lower      upper
## -0.5176824  0.9364907
```

```r
exp(confint(logFU))
```

```
##    lower    upper
## 0.595900 2.551013
```

```r
trio.augW <- augment.logFmatched(form,trio_dat,m=2)
logFW <- clogitlogF(form,trio.augW)
logFW
```

```
## Call:
## clogit(formula, dat, weights = dat$weights, method = "efron",
##     iter.max = maxit)
##
##     coef exp(coef) se(coef)     z     p
## g 0.1942    1.2143   0.3609 0.538 0.591
##
## Likelihood ratio test=0.29  on 1 df, p=0.5897
## n= 62, number of events= 29
```

```r
confint(logFW)
```

```
##      lower      upper
## -0.5124318  0.9171290
```

```r
exp(confint(logFW))
```

```
##     lower     upper
## 0.5990371 2.5020965
```

Session information for the trio data analysis

```
sessionInfo()
```

```
## R version 4.4.1 (2024-06-14)
## Platform: aarch64-apple-darwin20
## Running under: macOS Sonoma 14.7.8
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;  LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/Vancouver
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] lubridate_1.9.3 forcats_1.0.0   stringr_1.5.1   dplyr_1.2.0
##  [5] purrr_1.0.2     readr_2.1.5     tidyr_1.3.2     tibble_3.2.1
##  [9] ggplot2_4.0.2   tidyverse_2.0.0 coxphf_1.13.4   survival_3.6-4
##
## loaded via a namespace (and not attached):
##  [1] Matrix_1.7-0       gtable_0.3.6       compiler_4.4.1     tidyselect_1.2.1
##  [5] splines_4.4.1      scales_1.4.0       yaml_2.3.8         fastmap_1.2.0
##  [9] lattice_0.22-6     R6_2.5.1           generics_0.1.3     knitr_1.50
## [13] pillar_1.11.1      RColorBrewer_1.1-3 tzdb_0.5.0         rlang_1.1.7
## [17] stringi_1.8.4      xfun_0.52          S7_0.2.1           timechange_0.3.0
## [21] cli_3.6.4          withr_3.0.0        magrittr_2.0.3     digest_0.6.37
## [25] grid_4.4.1         rstudioapi_0.16.0  hms_1.1.3          lifecycle_1.0.5
## [29] vctrs_0.7.2        evaluate_1.0.1     glue_1.8.0         farver_2.1.2
## [33] rmarkdown_2.27     tools_4.4.1        pkgconfig_2.0.3    htmltools_0.5.8.1
```